

1. Process Synchronization and Child Process Monitoring Using waitpid()

Code:

```
priyanka@DESKTOP-HOISO7A: ~$ nano src/waitpid_demo.c
GNU nano 8.7.1 src/waitpid_demo.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("Child Process Started\n");
        printf("Child PID: %d\n", getpid());
        printf("Child Parent PID: %d\n", getppid());

        sleep(2);

        printf("Child Process Completed\n");

        exit(5);
    }

    // Parent process
    printf("Parent is waiting for Child PID: %d\n", pid);

    if (waitpid(pid, &status, 0) == -1)
    {
        perror("waitpid");
        return 1;
    }

    printf("\nParent Process Resumed\n");

    if (WIFEXITED(status))
    {
        printf("Child exited normally\n");
        printf("Child Exit Status: %d\n", WEXITSTATUS(status));
    }

    printf("Parent Process Completed\n");

    return 0;
}
```

Output:

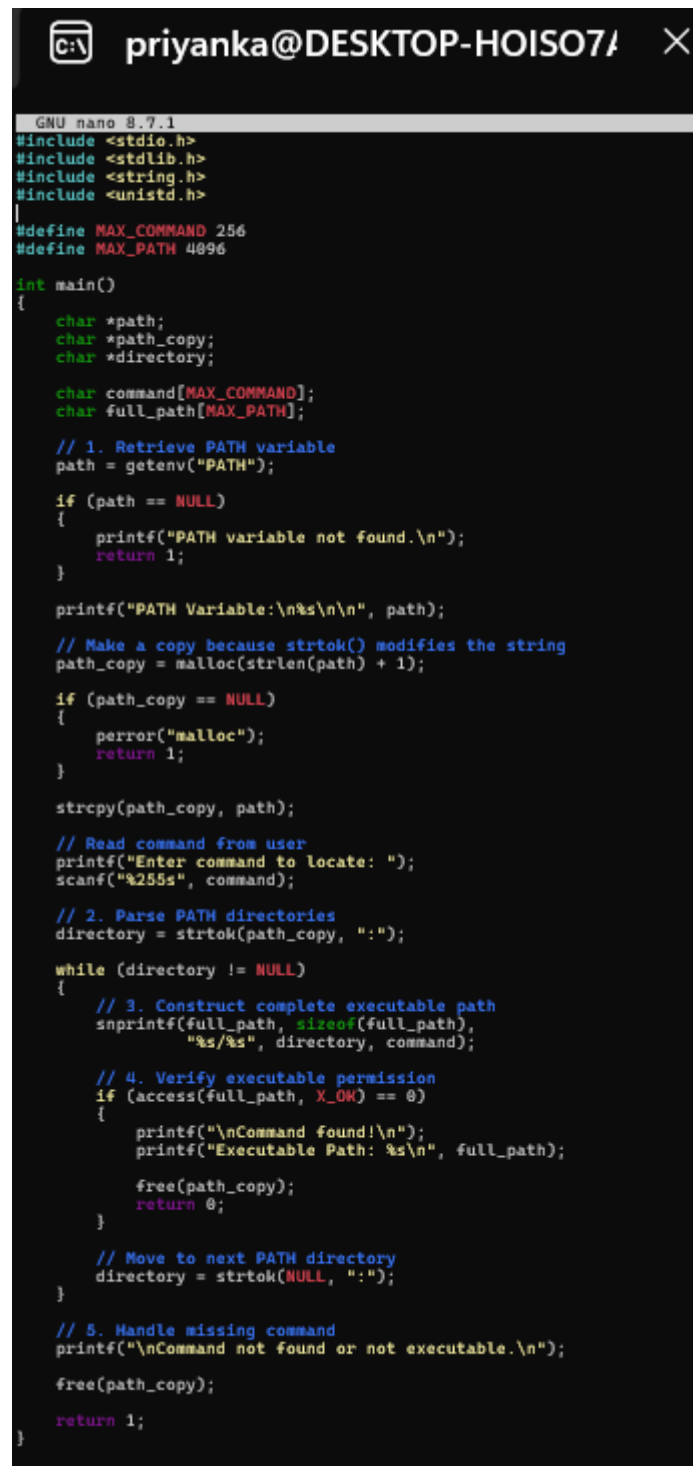
```
priyanka@DESKTOP-HOISO7A:~$ cd ~/OS_Skill
mkdir -p Week-4/src Week-4/bin
cd Week-4
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ nano src/waitpid_demo.c
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ gcc src/waitpid_demo.c -o bin/waitpid_demo
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ ./bin/waitpid_demo
Parent Process Started
Parent PID: 548

Parent is waiting for Child PID: 549
Child Process Started
Child PID: 549
Child Parent PID: 548
Child Process Completed

Parent Process Resumed
Child exited normally
Child Exit Status: 5
Parent Process Completed
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ |
```

2. To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

Code:



```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
|
#define MAX_COMMAND 256
#define MAX_PATH 4096

int main()
{
    char *path;
    char *path_copy;
    char *directory;

    char command[MAX_COMMAND];
    char full_path[MAX_PATH];

    // 1. Retrieve PATH variable
    path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH Variable:\n%s\n\n", path);

    // Make a copy because strtok() modifies the string
    path_copy = malloc(strlen(path) + 1);

    if (path_copy == NULL)
    {
        perror("malloc");
        return 1;
    }

    strcpy(path_copy, path);

    // Read command from user
    printf("Enter command to locate: ");
    scanf("%255s", command);

    // 2. Parse PATH directories
    directory = strtok(path_copy, ":");

    while (directory != NULL)
    {
        // 3. Construct complete executable path
        snprintf(full_path, sizeof(full_path),
                 "%s/%s", directory, command);

        // 4. Verify executable permission
        if (access(full_path, X_OK) == 0)
        {
            printf("\nCommand found!\n");
            printf("Executable Path: %s\n", full_path);

            free(path_copy);
            return 0;
        }

        // Move to next PATH directory
        directory = strtok(NULL, ":");
    }

    // 5. Handle missing command
    printf("\nCommand not found or not executable.\n");

    free(path_copy);

    return 1;
}
```

Output:

```
priyanka@DESKTOP-HOISO7A x + v
path_resolver.c waitpid_demo.c
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ gcc src/path_resolver.c -o bin/path_resolver
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ ls bin
path_resolver waitpid_demo
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ ./bin/path_resolver
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.10.7-hotspot/bin:/mnt/c/Maven/apache-maven-3.9.12/
bin:/mnt/c/Program Files/nodejs:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/
MySQL Shell 8.0/bin:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311/Scripts:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311:/mnt/c/Us
ers/DELL/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/DELL/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/DELL/AppData/Local/Python/bin:/mnt/
c/Users/DELL/AppData/Roaming/npm:/mnt/c/Users/DELL/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Program Files/MongoDB/Server/8.3/bin:/mnt/c/Program Fil
es/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311/Scripts:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311:/
/mnt/c/Users/DELL/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/DELL/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/DELL/AppData/Local/Python/
bin:/mnt/c/Users/DELL/AppData/Roaming/npm:/mnt/c/Users/DELL/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Users/DELL/mongosh-2.10.0-win32-x64/mongosh-2.
10.0-win32-x64/bin:/snap/bin

Enter command to locate: ls

Command found!
Executable Path: /usr/bin/ls
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ gcc src/path_resolver.c -o bin/path_resolver
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ ./bin/path_resolver
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.10.7-hotspot/bin:/mnt/c/Maven/apache-maven-3.9.12/
bin:/mnt/c/Program Files/nodejs:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/
MySQL Shell 8.0/bin:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311/Scripts:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311:/mnt/c/Us
ers/DELL/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/DELL/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/DELL/AppData/Local/Python/bin:/mnt/
c/Users/DELL/AppData/Roaming/npm:/mnt/c/Users/DELL/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Program Files/MongoDB/Server/8.3/bin:/mnt/c/Program Fil
es/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311/Scripts:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python311:/
/mnt/c/Users/DELL/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/DELL/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/DELL/AppData/Local/Python/
bin:/mnt/c/Users/DELL/AppData/Roaming/npm:/mnt/c/Users/DELL/AppData/Local/Programs/Antigravity IDE/bin:/mnt/c/Users/DELL/mongosh-2.10.0-win32-x64/mongosh-2.
10.0-win32-x64/bin:/snap/bin

Enter command to locate: gcc

Command found!
Executable Path: /usr/bin/gcc
priyanka@DESKTOP-HOISO7A:~/OS_Skill/Week-4$ |
```